Janith Ramanayake

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Personal Profile

A highly motivated Data Science graduate with a solid academic foundation in statistics, mathematics, and computer science, complemented by hands-on experience in machine learning and deep learning. I am currently seeking PhD opportunities in Data Science, with focused research interests in Signal Processing, Time Series Modeling, and Generative Adversarial Learning. Passionate about addressing real-world problems through data-driven, interdisciplinary approaches, and dedicated to advancing knowledge through collaborative research and impactful scientific contributions.

Education

BSc (Hons) in Data Science Apr 2021 – Apr 2025
University of Colombo | Faculty of Science GPA : 3.87|4 (First Class)

- **Statistics:** Probability and Distributions, Statistical Theory, Statistical Inference I/II, Multivariate Data Analysis, Time Series Analysis, Computational Statistics, Applied Statistical Models
- **Computer Science:** Introduction to Data Structures and Algorithms, Machine Learning I/II, Deep Learning, DataBase Systems I/II, Natural Language Processing, Image Analysis, Big Data Analytics, Fundamentals of Programming, Internet Technologies
- **Applied Mathematics:** Differential Equations I/II, Applied Linear Algebra, Computational Mathematics I/II, Linear Programming, Numerical Analysis, Optimization
- **Pure Mathematics:** Basic Analysis I/II/III, Introduction to Abstract Algebra, Introduction to Number Theory

Work Experience

Temporary Assistant Lecturer May 2025 – Present
Department of Statistics - University of Colombo Colombo, LK

- Assist in conducting practical laboratory and tutorial sessions for level 1 and 2 undergraduate courses:
  - IS 1007 - Introduction to Statistical Computing
  - ST 1008 - Probability & Distributions
  - ST 1009 - Exploratory Data Analysis
  - ST 2007 - Applications in Statistical Inference

Intern Data Scientist Nov 2024 – Apr 2025
Central Bank of Sri Lanka Colombo, LK

- Collaborated with senior statisticians and economists to integrate alternative data sources into macroeconomic analysis and policy research.
- Contributed to research and development of data-driven economic indicators using alternative data sources and machine learning techniques.
- Supported data visualization and reporting efforts by creating clear, informative charts and dashboards to communicate analytical insights to non-technical stakeholders.

Research Experience

Enhancing Sleep Stage Classification with Vision Transformers: A Study on Image Transformations, Data Augmentation, and Model Optimization Jun 2024 - Mar 2025

- Applied various time series to image transformation techniques (spectrograms, Gramian angular fields) to convert EEG signals into visual formats, evaluating their effectiveness compared to raw time series data for sleep stage classification.
- Utilized GAN-based methods to generate synthetic images from transformed EEG time series data, addressing class imbalance in sleep stages and enhancing subject-to-subject invariance in model learning.
- Employed pre trained Vision Transformer models with fine-tuning (ViT-B/16, ViT-B/32, DeiT-B, Swin-B,

PVT-Small) to evaluate classification performance and benchmark against conventional deep learning approaches.

A Data Driven Approach to Measuring Consumer Confidence in Sri Lanka with Google Search Trends and Deep Learning Jan 2025 - Mar 2025

- Collected and preprocessed high-frequency Google Trends data related to economic sentiment, consumption, and employment to serve as proxies for consumer behavior in Sri Lanka.
- Designed and trained deep learning models (e.g., LSTM, GRU, and hybrid architectures) to estimate a real-time Consumer Confidence Index, addressing limitations of discontinued survey-based indices.
- Evaluated model performance using historical economic indicators and validated index trends against major socio-economic events and policy changes.

Cross-Lingual Sentiment Modeling for Financial Texts in English, Sinhala, and Tamil Nov 2024 - Jan 2025

- Curated and annotated a trilingual financial text dataset from news articles, press releases, and social media in English, Sinhala, and Tamil to enable sentiment classification.
- Fine-tuned Llama 3.2 1B large language model for sentiment analysis across languages, focusing on domain adaptation to financial content.
- Evaluated model performance across languages using precision, recall, and F1-score, and addressed low resource challenges in Sinhala and Tamil through translation models.

Certifications

Deep Learning Specialization by DeepLearning.AI — Jan 2025

Machine Learning Specialization by DeepLearning.AI — Dec 2023

Technologies

Programming Languages: Python, R, Java, JavaScript

Mathematics and Statistics: Octave, SPSS, Minitab, Microsoft Excel

Frontend and Backend Development: React, React Native, Flask, FastAPI

Machine Learning and Data Science: TensorFlow, PyTorch, Power BI

Database Management: MySQL, MongoDB, BigQuery

Awards and Achievements

Honored on the Dean's List for academic excellence — 2021 - 2024

Awarded Faculty of Science Colours for Table Tennis — 2023 - 2024

Finalist, Intellihack NextGen Generative AI Hackathon organized by UCSC — 2024

Finalist, MiniHack Machine Learning Hackathon organized by Stat Circle, UoC FoS — 2023

Awarded Junior National Statistics Olympiad Merit Award. — 2015 - 2016

Extra Curricular Activities

Member of Stat Circle, Department of Statistics, University of Colombo — May 2023 - Apr 2025

Member of Table Tennis Club, University of Colombo — May 2021 - Apr 2024

Member of Leo Club, Faculty of Science, University of Colombo — May 2021 - Apr 2023

References

Dr. G A C N Priyadarshani
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Department of Statistics
University of Colombo
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Dr. H A S G Dharmarathne
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